

Representation Collapse Is Not the Bottleneck — and Neither Is Coverage or Sensing: A Negative Result and Capacity Diagnosis for Vision-Based Quadrotor Hover

Draft v0.6 — 2026-06-29. Target: ICRA / robot-learning workshop (venue TBD). **Simulation-only; no real-robot claim.** Every number is reproducible from the cited script/artifact. *Draft note (to be removed at submission):* v0.6 adds the **capacity scaling ladder (§6.5)** that directly tests the leading explanation left by v0.5: scaling `flow_net` $3.3\times$ (13.5 M $\rightarrow$ 44 M trainable, 3 seeds, frozen protocol) does **not** break the precision floor (best 2.63 m), so the robustness–precision conflict is more fundamental than parameter count; the Abstract, §1, §6.4, §7, §8 and §9 are re-synced (capacity changes from "leading untested explanation" to "tested, floor held"), and the ≈ 3 M parameter figure is corrected to the actual 13.5 M trainable. v0.5 added the decisive Teacher $\times$ Observation 2×2 (§6.4) and the scale-invariant-regulariser ablation (§6.1). The core Dispersive ablation (§5, official-code-faithful P2f: InfoNCE-L2 on the `flow_net` mid-block, $\lambda=0.5$, $\tau=0.5$, $/d$) is unchanged.

Abstract

Diffusion / flow-matching visual policies are prone to *representation collapse* when fine-tuned end-to-end, and **Dispersive Loss** — a contrastive regulariser that repels intermediate features — has been proposed to prevent it and thereby improve high-frequency visual control. We pre-registered and tested this hypothesis on a vision-based quadrotor hover task (monocular 64×64 FPV + IMU, flow-matching policy, 50 Hz closed loop). Under a **frozen evaluation protocol** (paired initial conditions, conditional-on-survival precision, bootstrap CIs, across-seed mean $\pm$ std, a *measured* state-based oracle), a **2×2 ablation** (Dispersive $\times$ end-to-end encoder, three seeds per cell) — run with an **official-code-faithful** Dispersive Loss (InfoNCE-L2 on the generative `flow_net` mid-block, $\lambda=0.5$, $\tau=0.5$, including the $/d$ per-dimension normalisation) — finds **no survival or task-precision gain above seed noise** (-2.2 pp Tier-1 within a 6.3 pp pooled std; survival -2.1 pp). Because the implementation follows the released code exactly — placement, weight, and per-dimension normalisation — the null cannot be attributed to a fidelity error. The frozen-encoder condition is an informative control rather than a no-op: under the faithful mid-block placement the term trains the generative network **even with a frozen encoder**, where it **destabilises** control (mean Tier-1 $87.8\rightarrow 74.4$, variance up) rather than leaving it inert. Reaching even this null verdict required a protocol hardened against **single-seed noise**: our own from-pixels PPO baseline swings between a 0 % and 47 % Tier-1 pass-rate on the training seed alone, so we report every retrained model as an **across-seed mean $\pm$ std** rather than a single leaderboard row. We then diagnose *why*. **(i)** Dispersive does not cure collapse — at **both** the faithful `flow_net` mid-block and an off-path placement (§6.1) it **games its own objective**, inflating feature norm ($\sim 9\times$ and $\sim 287\times$ respectively) while the intrinsic rank gets *worse*; replacing it with two **scale-invariant** criteria (unit-sphere InfoNCE, VICReg) removes the norm inflation and genuinely raises effective rank (from 3.5 % to 75–85 % of dimensions), yet still leaves closed-loop control flat — so survival is **decoupled** from feature geometry whatever criterion enforces dispersion. **(ii)** The only real survival mover is a transfer/conditioning/recovery **recipe**, not Dispersive. **(iii)** Precision is gated by **neither representation, sensing, nor coverage**. A renderer gate and a positive-control **intervention** rule out sensing: although the 64×64 FPV cannot encode metric range beyond ~ 2 m, that loss is a fixable renderer artifact (a non-saturating target restores it at 64 px), and *handing the policy the oracle metric position error barely moves precision* ($\sim 36\times$ the state oracle) while a richer cue **collapses survival**. We then build the competent far-range teacher and range-encoding observation this points to — a PID-CTBR teacher that recovers the 1–3 m band at 100 % survival (cond-IAE 0.14–0.18 m) and a perspective renderer that restores far-range image $\rightarrow$ distance R^2 $0.05\rightarrow 0.40$ — and remove **both** remaining candidate constraints jointly in a pre-registered three-seed **Teacher $\times$ Observation 2×2** under the same frozen protocol. The floor *holds* (cond-IAE 2.93 m, $\sim 43\times$ oracle, no better than the neither-factor control), and the two factors **negatively interact** on precision: coverage buys survival (+8–14 pp) but, stacked on the better-sensing cell, *worsens* precision ($2.48\rightarrow 2.93$ m). We conclude that for this class of task the binding constraint is none of representation collapse, sensing, or coverage, but a **robustness–precision conflict** — directly observed as this negative interaction. We then test its leading explanation, model capacity, with a third pre-registered ablation — a `flow_net` **scaling ladder** (13.5 M $\rightarrow$ 44 M trainable, 3 seeds): scaling the action backbone **$3.3\times$** does **not** break the floor (best 2.63 m, $\sim 39\times$ oracle; the gain small and saturating, with at most a within-noise survival cost), so the conflict is **more fundamental than parameter count** — leaving two distinct untested candidates we do not conflate: capacity *allocation* (a dedicated precision pathway) and a closed-loop / partial-observability limit. We release the protocol, all three ablations, the intervention, and all diagnostics.

1. Introduction

End-to-end visual control with generative policies (diffusion [3], flow matching [4]) has advanced rapidly, but a recurring failure mode is *representation collapse*: when the visual encoder is trained jointly with the action head, its features can degenerate onto a

low-dimensional subspace, losing the information the controller needs. **Dispersive Loss** [13] addresses this by adding a contrastive term that repels a batch's intermediate features, and has been reported to improve generative-policy control [14].

We asked a simple, falsifiable question on a concrete task — **vision-based quadrotor hover** — does Dispersive Loss improve closed-loop control, and does it need an end-to-end-trainable encoder to do so? The task is a good stress test: the policy runs at 50 Hz from a monocular 64×64 FPV stream plus IMU, with a flow-matching action head; covariate shift and feature collapse are both plausible failure modes.

Our answer is **no**, and the value of the paper is the *diagnosis* of why. We make three contributions:

1. **A frozen evaluation protocol** for visual hover that is robust to the short-survival *and* single-seed artifacts that silently mislead precision metrics and leaderboard rows for policies that crash early or train unstably (paired seeds, conditional-on-survival precision, bootstrap CIs, across-seed mean $\pm$ std, and a *measured* rather than assumed oracle).
2. **A pre-registered 2×2 falsification** (three seeds per cell) of the Dispersive-Loss hypothesis, hardened with an **official-code-faithful** implementation (InfoNCE-L2 on the `flow_net` mid-block, $\lambda=0.5, /d$), so the null result controls for implementation fidelity, plus a frozen-encoder control.
3. **A mechanistic diagnosis** that *excludes* all three candidate bottlenecks (representation, data coverage, sensing) rather than merely ranking them. A positive-control *intervention* hands the policy the oracle sensing signal and shows it does **not** restore precision (excluding sensing); a decisive, pre-registered **Teacher × Observation 2×2** then supplies a *competent* far-range teacher **and** a range-encoding observation jointly, and the precision floor still does not move (excluding coverage). What remains is a **robustness–precision conflict**, whose leading explanation — model capacity — we then test directly with a `flow_net` **scaling ladder** (13.5 M $\rightarrow$ 44 M trainable, 3 seeds): scaling the action backbone 3.3× does **not** break the floor, so the conflict is more fundamental than parameter count.

This is a negative result, but a constructive one: by excluding representation, sensing, and coverage in turn — and showing that a 3.3× larger action backbone does not break the floor either — it redirects effort toward capacity *allocation* (a dedicated precision pathway) and the closed-loop limit, not the auxiliary regularisers, richer sensors, larger backbones, or data-coverage fixes a reader would reach for first. Because the regulariser is implemented exactly as released — placement, weight, and per-dimension normalisation (§3) — the null result cannot be attributed to an implementation error.

2. Related Work

Dispersive regularization at a transfer boundary: from manipulation to high-speed visual hover.

Generative policies for visuomotor control. Behaviour cloning treats control as supervised regression from observations to actions, but robot action distributions are multimodal, sequentially correlated, and precision-sensitive, which makes plain regression brittle. Building on denoising diffusion [1] and its deterministic DDIM sampler [2], Diffusion Policy [3] reframed control as conditional denoising diffusion over short action sequences, with receding-horizon execution and a GroupNorm visual encoder trained end-to-end. Flow matching [4] and rectified flow [5] offer a continuous-time, simulation-free alternative that learns a velocity field from noise to data, since adopted as a lighter-weight policy backbone — e.g. the π_0 vision-language- action flow model [6] and the Riemannian flow-matching policy [7]. Our policy is a flow-matching action head over a fused vision–IMU conditioning vector, placing it in this family.

Because imitation alone cannot exceed its demonstrations, a second line fine-tunes pretrained generative policies with reinforcement learning. DPPO [8] casts the denoising chain as a multi-step MDP and applies PPO; ReinFlow [9] injects a learnable noise network into a flow policy's otherwise deterministic path, recovering a tractable likelihood that supports stable RL fine-tuning at very few — or even one — denoising steps; Q-score matching [12] is a related value-based route. This interactive, closed-loop setting should be distinguished from RL fine-tuning of *image-generation* diffusion models (DDPO [10]; DPOK [11]), which optimise a non-interactive sampler against a reward. This generative-policy-plus-RL stack is the substrate on which the regularizer studied here is layered, and the same stack in which the positive results for that regularizer were reported.

Representation regularization in generative models and policies. A recurring concern in denoising-based generators is that regression objectives supervise the output but leave intermediate features under-constrained, allowing them to collapse. The failure has a well-studied form: Jing et al. [22] show that even contrastive objectives suffer *dimensional collapse*, with embeddings spanning a lower-dimensional subspace than their nominal dimension — the property later diagnostics quantify through effective rank. REPA [15] addresses this by *aligning* a diffusion transformer's hidden states with a pretrained self-supervised encoder, at the cost of an external encoder and data; the classic variance/covariance and redundancy-reduction criteria of VICReg [16] and Barlow Twins [17] pursue the same anti-collapse goal without positive pairs. Dispersive Loss [13] removes the external-encoder cost differently: it keeps only the *repulsive* half of a contrastive objective, encouraging a generative network's intermediate representations to spread out with no positive pairs, no extra parameters, and no external data, validated on ImageNet class-conditional generation where it consistently improves FID.

Two robotics works carry Dispersive Loss into control. D²PPO [14] identifies "diffusion representation collapse" — semantically similar observations mapped to indistinguishable features — as a cause of manipulation failures, adds dispersive regularization treating every hidden representation in a batch as a negative pair, and reports average gains of 22.7 % in pre-training and 26.1 % after PPO fine-tuning on RoboMimic, with high real-robot success on a Franka Panda. DMPO [23] extends the idea to one-step MeanFlow policies, arguing dispersive regularization is necessary for stable single-step generation; concurrent MeanFlow-plus-dispersive manipulation work includes DM1 [18] and MP1 [19]. Notably, the public DMPO configuration specifies InfoNCE-L2 on the mid-block with weight 0.5 — the settings a faithful re-implementation in a new domain would inherit, and the settings we adopt in §3 / §5.

Three properties of this literature matter here. First, every validation is on image generation [13] or quasi-static, table-top manipulation scored by discrete task success [14, 23]. Second, the mechanism on offer is *observational discriminability*: the failure dispersive loss is claimed to fix is confusing two similar images. Third, none of these works publishes a fidelity audit — placement, weight, the per-dimension normalization in the official InfoNCE-L2 form — or a frozen-encoder control that isolates whether the term is even on the action-gradient path. Whether the benefit, the mechanism, or even the on-path behaviour transfers to a setting with a different bottleneck is left open.

Learning-based vision quadrotor control. High-speed visual flight is exactly such a setting. The dominant recipe couples privileged-learning imitation with sensor abstraction: Deep Drone Racing [24] maps images to a waypoint and desired speed for a downstream planner; Deep Drone Acrobatics [25] regresses collective thrust and body rates from abstracted visual-inertial inputs to fly 3g maneuvers by imitating a privileged expert; Learning High-Speed Flight in the Wild [20] maps depth and IMU to collision-free trajectories with zero-shot sim-to-real transfer; and champion-level drone racing was achieved with deep RL from onboard perception [21]. A consistent theme is that what unlocks robustness and transfer is *data diversity, sensor abstraction, and privileged supervision* — not auxiliary representation regularizers. The failure mode here is also different in kind: a hover or recovery policy does not fail by confusing two similar frames, it fails by drifting off-distribution and crashing, the classic covariate-shift problem of behaviour cloning [26].

Negative results, auxiliary-loss (in)effectiveness, and honest evaluation. Representation-centric add-ons do not help uniformly. Most directly, Schneider et al. [27] benchmark pretrained visual representations in model-based RL and find — "surprisingly" — that they are no more sample-efficient and no more OOD-robust than features learned from scratch, concluding that data diversity and architecture, not representation quality, drive OOD generalization. Establishing a credible negative result also depends on evaluation discipline: Henderson et al. [28] document how seed variance makes single-run RL claims unreliable; Agarwal et al. [29] show that few-run benchmarks are statistically fragile and advocate bootstrap intervals; Patterson et al. [30] catalogue maximization bias and unfair hyperparameter budgets. Together they motivate the frozen, multi-seed protocol with interval estimates (§4) on which our null effect is asserted rather than assumed.

Positioning: a transfer boundary, not a contradiction. We take Dispersive Loss following the official implementation — InfoNCE-L2 on the generative network's mid-block, weight 0.5, including the per-dimension normalization present in the released code — and ask whether the benefit reported for image generation [13] and table-top manipulation [14, 23] transfers to high-speed visual quadrotor hover. Under a frozen, three-seed protocol with bootstrap intervals (following [28, 29]), it does not: against D²PPO's reported +22.7 % / +26.1 % manipulation-success gains on RoboMimic [14], we measure a −2.2 pp Tier-1 change that sits inside a 6.3 pp across-seed std, with survival and precision unmoved (§5–§6). This is best read not as a contradiction of the manipulation results but as a *transfer boundary*. Where the bottleneck is observational discriminability — telling two near-identical grasp scenes apart — spreading intermediate features can help [14]. Where the bottleneck is behavioural coverage — recovering from a one-to-three-metre offset before crashing — feature dispersion addresses the wrong axis, echoing Schneider et al.'s [27] finding that representation quality is not the constraint governing out-of-distribution behaviour. Our contribution is a faithful, pre-registered, multi-seed demonstration of where a popular regularizer stops working on this task, and a diagnosis of why.

3. Task, Policy, and the Dispersive Mechanism

Task. Hover a quadrotor at a target. The simulator is a 6-DOF rigid body (200 Hz RK4 inner loop, INDI attitude control) with a 50 Hz outer control loop. The agent observes a 2-frame stack of 64×64 RGB synthetic FPV images and a 6-D IMU vector (gyro + specific force); it outputs a short sequence of collective-thrust/body-rate (CTBR) commands, executing the first and re-observing (receding horizon).

Policy (flow matching, v5). A CNN vision encoder produces a pooled feature $\text{vis_pooled} \in \mathbb{R}^{256}$ and a spatial map $\mathbb{R}^{\{256 \times 4 \times 4\}}$. An IMU MLP produces $\text{imu_feat} \in \mathbb{R}^{512}$. An IMU-to-vision cross-attention (IMU query, spatial keys/values) produces $\text{attended} \in \mathbb{R}^{256}$. The flow network conditions on $\text{global_cond} = [\text{attended}; \text{imu_feat}] \in \mathbb{R}^{768}$. A training-only auxiliary head predicts the 15-D privileged state from vis_pooled .

Dispersive Loss (faithful, official recipe). We apply Dispersive Loss exactly as released in the official code [13, 14]: the InfoNCE-L2 form on the generative network's intermediate block — here the `flow_net` mid-block (`return_mid=True`), `L_disp = log E_{i \neq j} [\exp(-D/\tau)]`, $D = \|\mathbf{z}_i - \mathbf{z}_j\|^2 / d$ — with weight $\lambda=0.5$ and $\tau=0.5$, including the per-dimension $/d$ normalization

present in the released code. (Implementation: `models/flow_policy_v5.py:204-224, 269-287`.) Because the term lands on `flow_net` mid-features, its gradient reaches the action path **regardless of whether the visual encoder is trainable** — the property that makes the frozen-encoder control in §5 informative.

Fidelity of the implementation, and an off-path variant studied in §6.1. Two configuration choices determine whether the term is active at all, and we fix both to the released values. *Placement, weight, and form*: the term must sit on a generative-network intermediate block (here the `flow_net` mid-features) at $\lambda=0.5$ in the InfoNCE-L2 form. A hand-rolled log-distance variant $-\text{mean}_{i \neq j} \log(\|x_i - x_j\| + \epsilon)$ at $\lambda=0.05$ applied to `vis_pooled` — a feature that feeds **only** the auxiliary state head, not the action path (which conditions on `attended` and `imu_feat`) — departs from the released recipe on all three axes. *Normalisation*: the published Algorithm 1 omits the per-dimension `/d` divisor present in the released code; without it the loss saturates to zero gradient at $\tau=0.5$, so this divisor is required for the term to carry gradient at all. The headline ablation (§5) uses the released configuration on both axes. Separately, §6.1 studies the off-path `vis_pooled` placement as a mechanistic probe, where its off-path position exposes how a scale-sensitive dispersion term can game its objective.

Oracle. A state-based PPO controller reading the privileged 15-D state is the performance upper bound.

4. Frozen Evaluation Protocol

The project's metric had changed three times (RMSE $\rightarrow$ linear-clip hierarchical $\rightarrow$ exp-decay hierarchical), making cross-run numbers incomparable; a **short-survival artifact** kept inflating precision for policies that crash early (a policy that dies at step 24 was credited with a "precise" 0.6 m hover from its few alive steps); and, as we demonstrate below with our own PPO baseline, **single-seed evaluation is itself unreliable** — a collapse-prone policy can swing from Tier-1 0 % to 47 % on the training seed alone. We freeze one protocol (`evaluate_frozen_p0.py`) and do not change it:

- **Paired initial conditions.** Episode i uses `seed = 12345 + i` to seed the env, the global RNG (visual domain randomisation), and torch (flow noise), so every model sees byte-identical initial conditions — a paired comparison.
- **Primary axes: survival and Tier-1 pass-rate** (fraction of 30 episodes flying $\geq 250/500$ steps). These are robust to the short-survival artifact.
- **Conditional precision.** Integrated absolute error (IAE) of position is reported **only over episodes that survived ≥ 250 steps** (`cond-IAE`, with its support `n_cond`); the naive all-episode IAE is reported but flagged.
- **Bootstrap 95% CIs** over episodes.
- **Across-seed mean $\pm$ std** for every model we *retrain* in this work — Table 1's P1 baselines and Table 2's 2×2 cells — over three training seeds. The legacy flow-policy checkpoints predate this work and are single-seed; that is a stated limitation, and the seed sensitivity we document below is exactly why a single training run is not a safe basis for a leaderboard row.
- **Measured oracle.** The state-based PPO oracle is rolled through the *same* protocol; its composite score (**0.9668**, 100 % survival, 0.068 m hover IAE) is the normaliser (`%oracle`), replacing a previously hard-coded constant.

Under this protocol, **three single-seed claims fail to survive**. Two come from the prior leaderboard: "H4 = SOTA" (true only on a multiplicative composite that rewards *die-early-but-precise*; H4 survives 40.8 %, Tier-1 13.3 %) and "RL gives 51 % precision gain" (a short-survival artifact: its conditional IAE is the *worst* of the group, 3.34 m, on `n_cond = 4/30`). The third we generate ourselves and catch with the seed protocol: on a single seed our PPO-from-pixels baseline "uniformly collapses" (Tier-1 0 %), but over three seeds it is only *collapse-prone* — a high-variance Tier-1 of **15.6 $\pm$ 22.0 %**, with one seed reaching 47 % — so the single-seed verdict simultaneously over- and under-states the behaviour. The lesson is structural, not incidental: on this task one training seed is not a safe basis for a leaderboard row, which is why every model we retrain is reported as a seed mean $\pm$ std. **No vision policy exceeds ~ 17 % of oracle; none is deployable.** (Table 1.)

Table 1 — Frozen leaderboard (flow policies: `evaluation_results/frozen_p0_leaderboard.json`; P1 baselines ^B, 3 seeds: `evaluation_results/baselines_frozen_leaderboard.json`, `evaluation_results/baselines_frozen_ppopx.json`, per-seed `..._{bc,ppopx}_s{1,2}.json`, aggregate `...seeds_aggregate.json`).

Model	Score	Survive	Tier-1	cond-IAE (n)	%Oracle
PPO Oracle (state)	0.967 [0.966, 0.967]	100 %	100 %	0.068 m (30)	100 %
H4_BC	0.168 [0.154, 0.182]	40.8 %	13.3 %	2.520 m (4)Δ	17.4 %
v5_RL_best	0.129 [0.117, 0.139]	28.8 %	13.3 %	3.340 m (4)Δ	13.3 %
v5_BC	0.124 [0.107, 0.143]	55.9 %	70.0 %	2.724 m (21)	12.8 %
PPO-from-pixels ^B	0.114 $\pm$ 0.017	30.1 $\pm$ 15.6 %	15.6 $\pm$ 22.0 %	2.83 m (1/3)ΔΔ	11.8 $\pm$ 1.8 %
Joint_E2E_v5	0.106 [0.095, 0.118]	62.2 %	80.0 %	3.077 m (24)	11.0 %
BC-vision-only ^B	0.089 $\pm$ 0.018	54.7 $\pm$ 3.4 %	61.1 $\pm$ 13.4 %	3.30 $\pm$ 0.36 m (18)	9.2 $\pm$ 1.8 %

△ $n_{\text{cond}} = 4$: cond-IAE unreliable; the low all-IAE is a short-survival artifact. △△ PPO-from-pixels is collapse-prone, not uniformly collapsed: 2 of 3 seeds survive 0/30 past 250 steps ($n_{\text{cond}} = 0$, cond-IAE undefined; their low all-IAE 0.70–0.87 m is a pure short-survival artifact, crashing ~75 steps), while 1 seed reaches Tier-1 46.7 % (cond-IAE 2.83 m). The cond-IAE shown is that single surviving seed (1/3). ^B P1 baseline (this work): **mean $\pm$ std over 3 seeds {0,1,2}** (population std, matching Table 2); the bracketed [.,.] on the other rows are bootstrap 95 % CIs over a single checkpoint's episodes. BC-vision-only = no IMU, no flow (plain vision→action MLP); PPO-from-pixels = end-to-end RL from pixels (1 M steps, CTBR).

The two P1 baselines (3 seeds each) sit within the flow-policy band and reinforce the diagnosis. A plain vision→action regressor (BC-vision-only: no IMU, no flow) already reaches **9.2 $\pm$ 1.8 % of oracle at Tier-1 61.1 $\pm$ 13.4 %** — *on par with* the IMU-fused flow policies v5_BC (12.8 %, Tier-1 70 %) and Joint_E2E_v5 (11.0 %, Tier-1 80 %), well within the wide across-seed spread. Flow modelling and IMU fusion thus buy little in survival on this task: **policy sophistication is not the bottleneck**. The other baseline, PPO-from-pixels, is the collapse-prone case dissected above; here it doubles as a clean short-survival-artifact demonstration. On its two collapsed seeds the deceptively low *all*-IAE (0.70–0.87 m) is a textbook instance of what the §4 protocol is built to expose — they "look precise" only because they crash (~75 steps) before they can drift — whereas averaging over seeds raises the all-IAE to 1.41 $\pm$ 0.89 m once the surviving seed's drift is counted. Both baselines stay far below the 17 % ceiling; none is deployable.

One training seed is not a safe leaderboard row (P1 baselines, 3 seeds each)

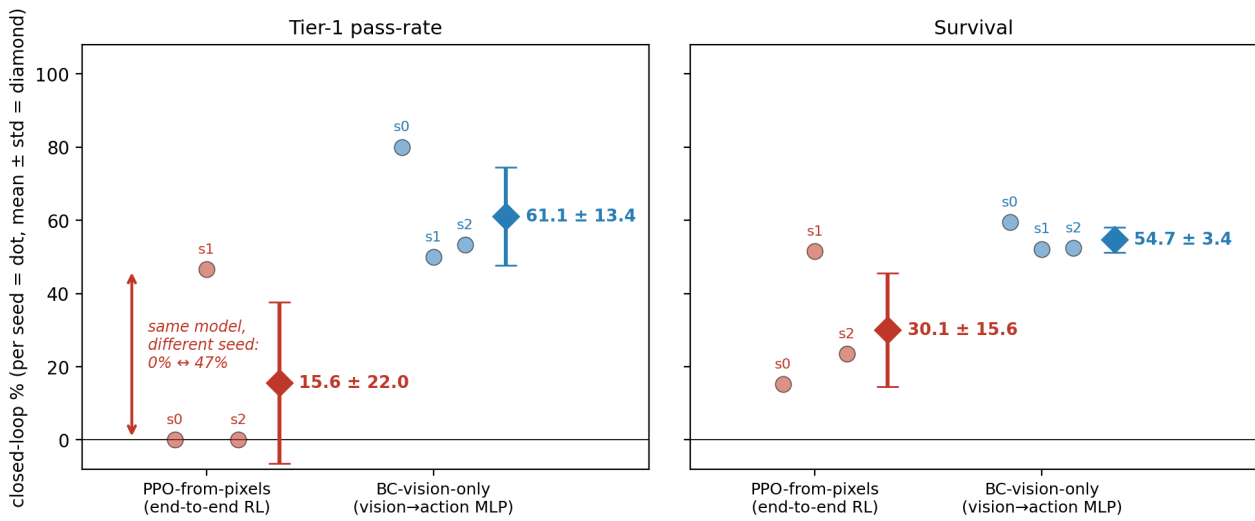


Figure 1. Per-seed Tier-1 pass-rate (left) and survival (right) for the two P1 baselines, three training seeds each (dots, labelled s0–s2), with the across-seed mean $\pm$ std (diamond). PPO-from-pixels swings from 0 % (seeds 0, 2 — would read as "uniformly collapsed") to 47 % (seed 1) on Tier-1 from the training seed alone (15.6 $\pm$ 22.0 %), so any single-seed leaderboard row simultaneously over- and under-states it; BC-vision-only is comparatively stable (61.1 $\pm$ 13.4 %). This seed sensitivity is why every retrained model in this work is reported as a three-seed mean $\pm$ std (`scripts/make_paper_figures.py`, `evaluation_results/baselines_frozen_seeds_aggregate.json`).

5. The Core Ablation: Dispersive $\times$ E2E (2 $\times$ 2, 3 seeds)

We vary two factors, holding everything else fixed (H4-transfer init, 500 hover + 500 recovery demos, 80 epochs, lr 1e-4, batch 256, task-conditioned):

- **Dispersive**: the official-code-faithful InfoNCE-L2 term (§3) at $\lambda=0.5$ (ON) vs 0.0 (OFF), on the `flow_net` mid-block.
- **E2E**: vision encoder trainable (ON) vs frozen at the transferred init (OFF).

Design and power. With three seeds per cell and a pooled across-seed Tier-1 std of ~6.3 pp, this design is powered to detect only large effects (≥ 6 pp Tier-1); a smaller true effect of Dispersive cannot be excluded. We therefore report a null in the sense of *no effect above seed noise*, not a proof of exact zero, and judge support by whether D1E1 exceeds DoE1 by more than the pooled std.

What the frozen-encoder cell tests. Because the term lands on `flow_net` mid-features, it trains the generative network *even with a frozen encoder*. The frozen-encoder cell (Eo) is therefore not a trivial no-op: it isolates Dispersive's effect with the *encoder* held fixed but the *flow network* still adapting. The decisive hypothesis test is **D1E1 vs DoE1** (Dispersive ON vs OFF, both E2E).

Table 2 — 2 $\times$ 2 ablation, faithful Dispersive (`evaluate_p2_ablation.py`, 3 seeds, frozen protocol; `evaluation_results/p2f_ablation_leaderboard.json`).

Cell	Disp / E2E	Tier-1 (mean $\pm$ std)	Survival (mean $\pm$ std)	cond-IAE	%Oracle
DoEo	OFF / frozen	87.8 $\pm$ 3.1	66.1 $\pm$ 3.7	2.93 m	13.4 %
D1Eo	ON / frozen	74.4 $\pm$ 8.7	60.4 $\pm$ 1.6	2.81 m	12.9 %
DoE1	OFF / E2E	92.2 $\pm$ 3.1	65.0 $\pm$ 2.8	2.91 m	12.4 %
D1E1	ON / E2E	90.0 $\pm$ 5.4	62.9 $\pm$ 2.4	2.89 m	12.2 %

Faithful Dispersive $\times$ E2E (2 $\times$ 2, 3 seeds): no Dispersive effect above seed noise

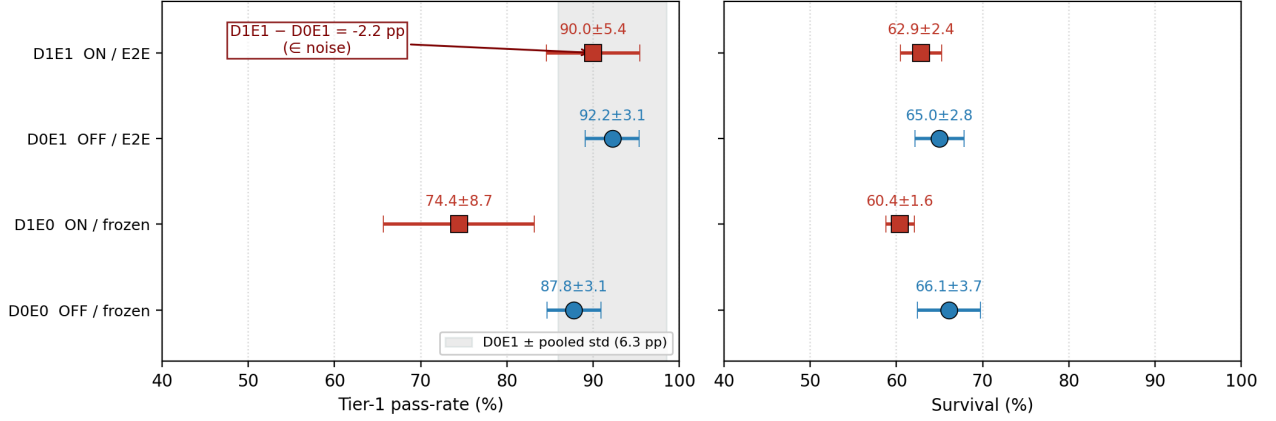


Figure 2. The 2 $\times$ 2 ablation as a forest plot (faithful Dispersive, 3 seeds, mean $\pm$ across-seed std). *Left:* Tier-1; the grey band is DoE1 $\pm$ the pooled across-seed std (6.3 pp). The decisive D1E1 (ON/E2E) sits inside it — **no Dispersive effect above seed noise** (-2.2 pp). The one cell that moves is D1Eo (ON/frozen, 74.4 ± 8.7), where Dispersive-on-frozen lowers the mean and inflates the variance. *Right:* survival is flat across all four cells. (scripts/make_paper_figures.py $\rightarrow$ evaluation_results/p2f_ablation_leaderboard.json.)

Findings. - **Dispersive is not supported (Figure 2).** D1E1 vs D0E1: Tier-1 -2.2 pp inside the 6.3 pp pooled across-seed std; survival -2.1 pp; conditional precision unchanged (2.89 vs 2.91 m). The decision rule (supported iff D1E1 beats D0E1 by more than the pooled std) returns **NOT supported**. The conclusion holds under the off-path, low-weight variant of §6.1 as well ($+1.1$ pp), so it does not hinge on placement or weight. - **The frozen-encoder condition is an informative control, not a no-op.** Under the faithful mid-block placement the encoder-frozen runs *do train flow_net*, so the checkpoints differ (`p2f_D1E0_s*  $\neq$  p2f_D0E0_s*`, MD5 distinct across all three seeds). The frozen row is not inert: Dispersive-on-frozen **destabilises** control (mean Tier-1 $87.8 \rightarrow 74.4$, variance $3.1 \rightarrow 8.7$ pp) — with no trainable encoder to co-adapt, repelling the mid-features pushes the mean down and the variance up rather than helping. - **E2E is the only (small) Tier-1 mover** (DoEo $\rightarrow$ DoE1, $+4.4$ pp), and even it neither extends survival nor improves precision (§6.2).

All four cells beat the prior frontier Joint_E2E_v5 (Tier-1 80 %), but §6.2 shows that gain is the *recipe*, not Dispersive. Precision is unmoved (cond-IAE 2.7–3.2 m, ~ 12 –13 % oracle): **nothing is deployable**. This same 2 $\times$ 2 methodology — frozen Po, three seeds per cell, the pooled-std decision rule — is re-used unchanged for the second, precision-focused ablation in §6.4 (Teacher $\times$ Observation), so the two falsification tests rest on one protocol; only the manipulated factors and the primary axis (there, cond-IAE) differ, as each hypothesis dictates.

6. Diagnosis: Which Bottleneck?

Three candidate explanations for the flat result: representation (the thing Dispersive targets), data coverage, or sensing. We test the first two with observation-only measurements (no retraining), and the third with both a renderer gate and a positive-control **intervention** that hands the policy the very signal it is presumed to lack (§6.3).

6.1 Representation: Dispersive games its objective; survival $\perp$ rank

Placement note. We measure feature geometry at **both** placements. We start with the off-path `vis_pooled` placement (Table 3; legacy P2 checkpoints): because `vis_pooled` feeds only the auxiliary state head, it cleanly isolates how a scale-sensitive dispersion term games its objective off the action path. We then confirm the same pathology at the faithful `flow_net` mid-block, where the official term actually acts (Table 4; the p2f checkpoints of §5). One placement-dependent detail: off-path, a frozen encoder zeroes the gradient (so D1Eo $\equiv$ DoEo, the legacy no-op); at the faithful placement the term trains `flow_net` even when frozen (§5), so D1Eo $\neq$ DoEo.

We push a fixed 4000-image hover+recovery batch through each checkpoint's encoder and measure the geometry of `vis_pooled` (D=256). (`measure_feature_collapse.py`, legacy P2 checkpoints.)

Table 3 — feature geometry (mean over 3 seeds).

Cell	eff_rank	dims for 99% var	feat_norm	L_disp	mean cos
DoEo / D1Eo (frozen)	30.3	91	1.80	-0.81	0.071
DoE1 (E2E)	9.0	45	11.42	-1.23	0.956
D1E1 (E2E + Disp)	2.0	2	3281	-8.14	0.013

Three things follow. **(i)** The frozen no-op is confirmed at the feature level ($D1Eo \equiv DoEo$ to 4 d.p.). **(ii)** Collapse is *real*: naive E2E ($DoE1$) drops effective rank 30→9 and drives mean pairwise cosine to 0.96 (features nearly colinear) — there *was* something for Dispersive to fix. **(iii)** Dispersive does not fix it; it **games its objective**. It minimises L_{disp} (-1.23→-8.14) and inflates pairwise distance $\sim 1270\times$, but does so purely by inflating the feature norm $\sim 287\times$ (11.4→3281); the *intrinsic* dimensionality gets **worse** (effective rank 9→2; 99.8 % of variance on 2 dims). The "healthy-looking" low cosine (0.013) is a 2-D-at-huge-radius artifact, not high-dimensional spread. Because `vis_pooled` feeds only the auxiliary head (§3), the cheapest way to satisfy the distance objective is to inflate the pooled-specific projection, leaving the action-relevant spatial pathway comparatively untouched.

The decisive plot is rank vs survival (Figure 3). Effective rank swings $15\times$ (30→9→2) while closed-loop survival stays flat (66.1→65.0→65.0 %, within ~ 3 pp std). **Survival is decoupled from `vis_pooled` rank.** Within this off-path probe, neither preventing collapse ($DoEo$ rank 30) nor worsening it ($D1E1$ rank 2) moves the closed-loop outcome. Together with the faithful on-path null (§5), this is the paper's strongest evidence that, *for this task*, representation collapse is not the binding constraint. The same pathology and decoupling hold at the faithful placement (Table 4, below).

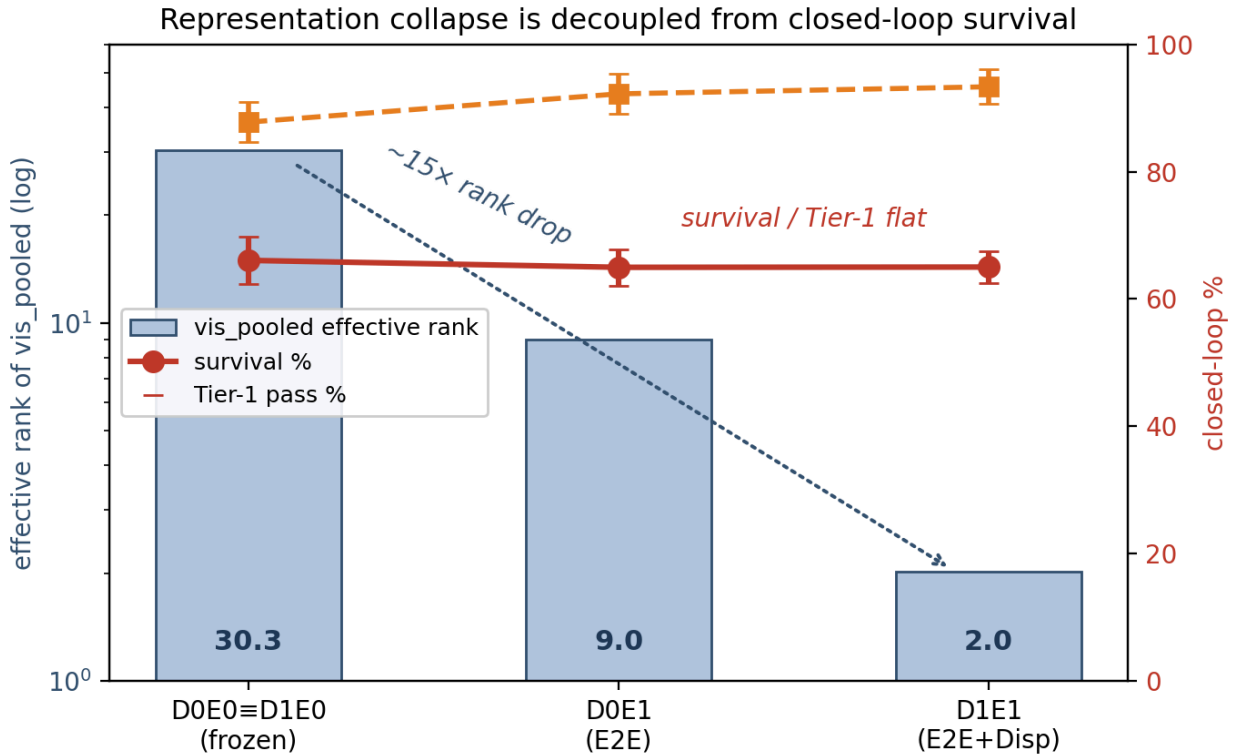


Figure 3. `vis_pooled` effective rank (bars, log axis) collapses $\sim 15\times$ across the 2×2 cells (30.3 frozen → 9.0 E2E → 2.0 E2E+Dispersive), while closed-loop survival and Tier-1 pass-rate (lines, right axis; error bars = across-seed std) stay flat. Survival is decoupled from the representation geometry Dispersive targets.

The same pathology holds at the faithful placement (Table 4). The off-path probe above cannot say whether the faithful `flow_net` mid-features — where the official term actually acts — behave the same way. Measuring them directly (same fixed 4000-sample batch, now of (image, IMU, action, timestep t)) so the mid-features are well defined; `measure_feature_collapse_flowmid.py`) returns the same verdict, more sharply. Dispersive ON drives the InfoNCE-L2 objective to its subset floor ($\log(1/2048) = -7.62$, identical across all three seeds) — it fully *wins* its objective — but does so purely by inflating the mid-feature norm $\sim 9\times$ (9.4→84) while the effective rank **collapses** (221→36 with a trainable encoder, 249→37 frozen; dims-for-99 %-variance 683→177). Exactly as off-path, the term games a scale-sensitive objective by norm growth and *worsens* intrinsic rank. Two further points: (i) naive E2E barely dents the mid-feature rank ($DoEo \rightarrow DoE1$, 249→221) — unlike `vis_pooled` (30→9), the action-path mid-features do not strongly collapse on their own, so there is little for Dispersive to "fix"; (ii) the frozen-Dispersive cell ($D1Eo$) shows the *same* norm blow-up and rank collapse as $D1E1$, a plausible mechanism for its closed-loop destabilisation (§5) — repelling the velocity field's own mid-features inflates them $\sim 9\times$ with no encoder free to compensate. Across all four cells the

mid-feature rank swings $\sim 7\times$ (249 $\rightarrow$ 36) while survival stays within ~ 3 pp (§5): the rank-vs-survival decoupling holds at the faithful placement too.

Table 4 — faithful `flow_net` mid-block geometry (D=1024, mean $\pm$ std over 3 seeds; `measure_feature_collapse_flowmid.py` $\rightarrow$ `evaluation_results/p2f_feature_collapse_flowmid.json`).

Cell	eff_rank	dims for 99% var	feat_norm	disp_infonce	mean cos
DoEo (OFF / frozen)	248.8 $\pm$ 9.0	722	9.35	-0.283	0.174
D1Eo (ON / frozen)	36.7 $\pm$ 0.5	189	83.8	-7.624	0.058
DoE1 (OFF / E2E)	221.4 $\pm$ 16.2	683	9.49	-0.293	0.163
D1E1 (ON / E2E)	35.9 $\pm$ 0.6	177	84.8	-7.624	0.056

Scale-invariant criteria cure the collapse but still do not move control (Table 5). One escape remains for the faithful term: perhaps it fails only because its *scale-sensitive* L2-InfoNCE objective is gameable by norm inflation, and a criterion that cannot inflate its way out would both produce genuine dispersion *and* help control. We test this directly, holding the D1E1 recipe fixed and varying **only** the dispersion criterion on the same `flow_net` mid-block (3 seeds each, frozen protocol; `run_p6_form_ablation.py`): a **unit-sphere InfoNCE** (cosine; norm inflation impossible by construction) and a **VICReg-style** variance + covariance term. Both behave exactly as intended on the geometry — feature norm stays O(1) (1.33 $\times$ / 1.36 $\times$ the no-dispersion baseline, against the faithful term's 8.93 $\times$) and effective rank *rises* to **75 % / 85 %** of the 1024 dimensions (against the faithful term's collapse to 3.5 %): they achieve the high-rank dispersion the regulariser is designed for, without the cheat. Yet closed-loop control does not improve — all forms sit in one band (survival 60–65 %, Tier-1 82–92 %, cond-IAE 2.9–3.1 m), and the only metric to cross the pooled seed std does so in the *worse* direction (cond-IAE +0.13–0.21 m). A **24 $\times$ swing in mid-block effective rank (36 $\rightarrow$ 867) buys $\approx$ 0 control**. This closes the "you used a scale-sensitive criterion" escape and is the on-path counterpart of the `vis_pooled` decoupling above: representation geometry is decoupled from closed-loop control *regardless of which criterion enforces dispersion*.

Table 5 — scale-invariant dispersion forms (D1E1 recipe, only the criterion varies; 3 seeds, geometry from `measure_feature_collapse_flowmid.py`, control from frozen PO; `evaluation_results/p6_form_ablation_leaderboard.json`, `docs/experiment_report_p6_scale_invariant.md`).

Form on <code>flow_net</code> mid-block	eff_rank (of 1024)	feat_norm $\div$ off	Survival	Tier-1	cond-IAE
off (DoE1, no dispersion)	221 (22 %)	1.00 $\times$	65.0 $\pm$ 2.8	92.2 $\pm$ 3.1	2.91 m
InfoNCE-L2 (faithful, D1E1)	36 (3.5 %)	8.93$\times$	62.9 $\pm$ 2.4	90.0 $\pm$ 5.4	2.89 m
unit-sphere InfoNCE (cosine)	769 (75 %)	1.33 $\times$	64.4 $\pm$ 3.3	86.7 $\pm$ 8.2	3.10 m
VICReg (var + cov)	867 (85 %)	1.36 $\times$	60.5 $\pm$ 2.4	82.2 $\pm$ 11.0	3.03 m

6.2 The only survival mover is the recipe, not Dispersive (or E2E)

Decomposing the gain over the prior frontier as controlled steps (`experiment_report_survival_movers.md`):

Step	Δ Tier-1	Δ Survival	cond-IAE
prior frontier Joint_E2E_v5 $\rightarrow$ recipe (frozen cell DoEo)	+7.8 pp	+3.9 pp	2.93 m
+ E2E (DoEo $\rightarrow$ DoE1)	+4.4 pp	-1.1 pp	2.91 m
+ Dispersive, faithful (DoE1 $\rightarrow$ D1E1)	-2.2 pp ($\in$ noise)	-2.1 pp	2.89 m

(The DoEo/DoE1 cells are Dispersive-OFF, so the +Dispersive step is the only one that depends on the regulariser: under the official configuration it is -2.2 pp, and under the off-path low-weight variant of §6.1 it is +1.1 pp — both inside the across-seed std.)

The dominant lift is the **recipe** (H4-transfer init + task-conditioning + recovery mix), and it holds with a **frozen** encoder — so it is not from E2E. E2E adds a small Tier-1-only bump that neither extends survival nor improves precision. Dispersive adds nothing (slightly negative under the official configuration, $\in$ noise). **Mover ranking: recipe $\gg$ E2E $\gg$ Dispersive $\approx$ 0.**

6.3 Precision is not sensing-gated — leaving a coverage hypothesis to test

cond-IAE is ~ 2.8 m (~ 13 % oracle) for *every* configuration. Why won't precision move? This subsection rules out **sensing** with a renderer gate and a positive-control intervention, leaving a single leading candidate — a coverage / teacher-competence gap — which §6.4 then tests directly and also excludes.

Coverage probe (`measure_ood_coverage.py`). Comparing position-error magnitude in the BC data vs the closed-loop steady state: training is overwhelmingly at-target (95.8 % of timesteps within 0.2 m; p99 = 0.62 m), while the surviving policy spends ~ 90 % of its steady state beyond 1 m and 46 % beyond 3 m. **97.3 %** of steady-state samples lie above the training p99; **25 %** above the

training maximum (zero-coverage). The policy lives where it has ~no labels — a real coverage gap, *consistent* with either a data limit or a sensing limit.

Is the coverage gap data-fixable, and does the FPV even encode range? Three observation-only checks bear on it: 1. **Widening the recovery init does not change coverage.** The env anchors `target = init_pos` for hover, so a wider `--pos-range` just relocates the hover point; a freshly collected ± 3 m dataset has *identical* position-error coverage to the ± 1 m one (both 0.4 % of timesteps beyond 1 m). 2. **The expert itself cannot recover from > 2 m offset** (`target  $\neq$  init`): 1 m and 2 m recover to 0.066 m (20/20), but 3 m crashes (0/20). Even a correct offset collection tops out at ~ 2 m — below the precision-limiting regime. 3. **The FPV image does not encode metric range** (`measure_image_distance_info.py`; **Figure 4**). The only range cue is the target crosshair size, `size = max(2, min(6, [6/(d+0.5)] + dr))`. With domain randomisation, adjacent-distance separability is $d' < 0.2$ everywhere, and a linear decode of distance from the image has $R^2 = 0.41$ near (< 1 m) but only **0.12 far** (≥ 1.5 m). With DR off (noiseless ceiling) the size **saturates at 2 px for all $d \geq 2$ m**, so renders at 2.0, 2.5, and 3.0 m are **byte-identical** ($d' = 0$). The crosshair *position* encodes only normalised direction (range-invariant).

The policy's 2.83 m steady drift sits exactly where the observation carries **no recoverable metric range** (Figure 4). One might conclude that precision is *information-gated by the observation model*. That inference does not follow: a measurement that the FPV does not encode range does not establish that the missing range is the *binding* cause of imprecision. We test that implication directly with a renderer gate and a positive-control intervention, and **both refute it**.

Measurement: the FPV cannot encode metric range where the policy operates (but see Fig. 5 — this is a fixable artifact, and range is not what gates precision)

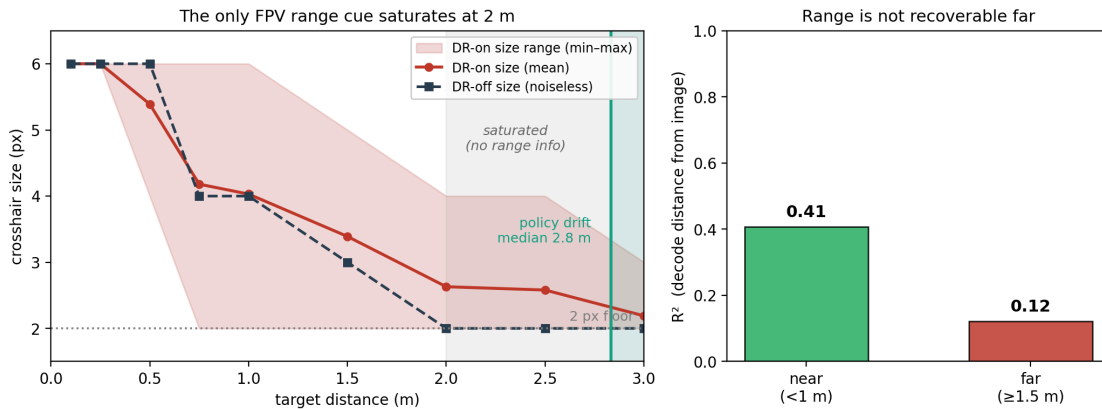


Figure 4. *Left:* the only FPV range cue (target crosshair size) saturates at the 2 px floor beyond ~ 2 m — under domain randomisation (red band = min–max) the size is uninformative even nearer, and with DR off (dashed) renders at 2.0/2.5/3.0 m are byte-identical. The policy's steady-state drift (green, median 2.83 m) sits inside the saturated region. *Right:* a linear decode of distance from the image succeeds near ($R^2=0.41$, < 1 m) but fails far ($R^2=0.12$, ≥ 1.5 m). This is a *measurement* of the observation, not the binding cause of imprecision (Figure 5).

The information loss is a renderer artifact, not the pixel count (gate). Before treating "richer sensing" as the fix, we ask whether a more capable monocular renderer would even *carry* the far-range information (`measure_higher_res_gate.py`; **Figure 5, left**). Decoding distance from the image across $\{64, 128, 256\}$ px $\times$ {production saturating crosshair, perspective non-saturating target}, the saturating crosshair's far-range R^2 is ≈ 0 at *every* resolution ($-0.01 / 0.11 / 0.04$) — raising resolution alone does nothing — whereas a perspective (optical-expansion) target lifts far R^2 to **0.42 at the same 64 px** ($0.50 / 0.45$ at 128 / 256 px). So the §6.3 measurement is partly an artifact of this synthetic renderer's quantised crosshair, fixable without any resolution change; even so, the perspective target's far R^2 (0.45) $\ll$ its near R^2 (0.88), so a richer sensor would *improve*, not solve, far-range precision.

Even the oracle range cue does not move precision (intervention). The decisive test supplies the missing channel directly: we fold the metric body-frame position error the FPV cannot encode (`states[:, :3]`, computed from the existing data — no re-collection) into the policy's conditioning, retrain the DoE1 frontier recipe unchanged, and re-eval under the frozen protocol (3 seeds; control = the no-cue DoE1; **Figure 5, right**). Handed the *oracle* metric range (scalar $\|pos\text{-}err\|$, $\sigma=0$), cond-IAE moves only $2.91 \rightarrow 2.43$ m — still $\sim 36\times$ the 0.068 m state oracle, and bought with -6.7 pp survival / -13 pp Tier-1. A realisable sensor ($\sigma=0.15$ m) erases even that (2.81 m $\approx$ control). The *richer* full 3-D position cue is actively harmful: it **collapses survival across all three seeds** (Tier-1 $92 \rightarrow 7\%$, surviving to the 250-step threshold in $\approx 2/30$ episodes), so its conditional precision is an artifact, not a win. Supplying range tells the drone *how far off* it is, but not *what to do* about it — and no demonstration in the 1–3 m band teaches that.

Verdict: precision is not sensing-gated. Range information — even oracle, even full position — does not restore precision, and a richer cue harms survival. That leaves a single leading candidate, which the measurements above motivate but do **not** prove: the absence of a learned far-range *recovery behaviour* in the 1–3 m band. The BC data has no labels there (the coverage gap above) and the *current* state-PPO expert cannot generate them (it crashes from > 2 m, check 2). The natural inference — *precision is coverage*

/ *teacher-competence-gated*, and would move given a competent far-range teacher — is exactly the kind of "fixable-bottleneck" conclusion that a negative-result paper should not assert from observational probes alone. **We therefore do not stop here.** §6.4 builds the very thing this inference calls for — a teacher competent across the full 1–4 m band *and* the coverage and range-encoding observation it implies — and tests, decisively, whether precision then moves. (It does not: the coverage hypothesis is excluded, leaving a capacity conflict.)

Precision is coverage/teacher-competence-gated, not sensing-gated

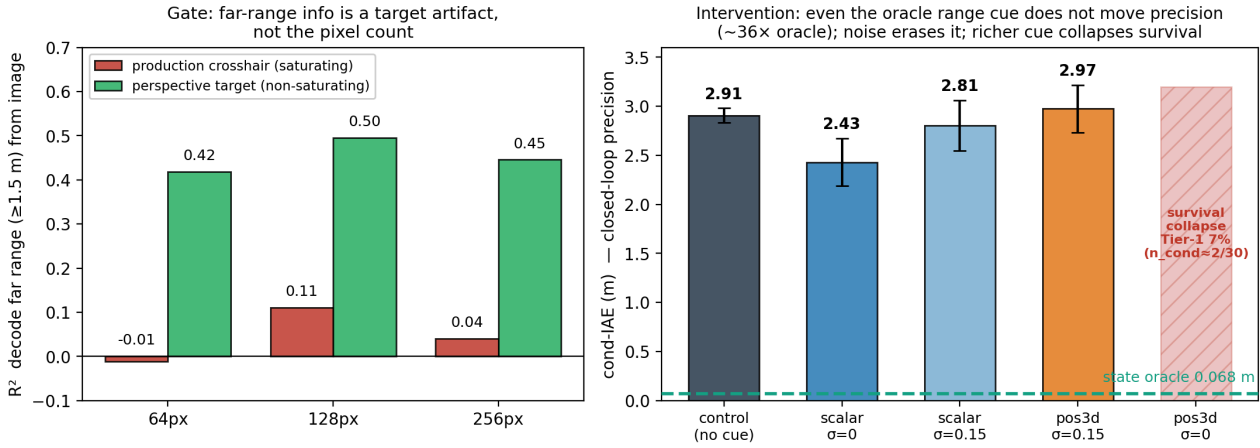


Figure 5. *Left (gate):* far-range (≥ 1.5 m) image→distance R^2 for three resolutions $\times$ two target renderers. The saturating production crosshair is uninformative at every resolution (raising pixels does nothing); a non-saturating perspective target restores ~half the far-range information already at 64 px — the loss is a target artifact, not the pixel count. *Right (intervention):* closed-loop precision (cond-IAE, 3 seeds) when the policy is *handed* the metric position error the FPV lacks. Even the oracle scalar range ($\sigma=0$) leaves precision $\sim 36\times$ the state oracle (green dashed, 0.068 m); sensor noise erases the gain; the richer 3-D cue ($\sigma=0$) collapses survival (hatched; conditional IAE unreliable). Precision is **not sensing-gated**; the coverage / teacher-competence hypothesis this points to is built and tested decisively — and also excluded — in §6.4.

6.4 The decisive test: a competent teacher and a range-encoding observation still do not break the floor (Teacher $\times$ Observation 2 $\times$ 2)

§6.1 excluded representation and §6.3 excluded sensing, leaving one leading hypothesis: precision is **coverage / teacher-competence-gated** and would move given a teacher competent in the 1–3 m band. A negative-result paper should not rest on that inference from observational probes — so we *build the fix the hypothesis demands* and test it under the same frozen protocol. This mirrors §5: a pre-registered 2 $\times$ 2, three seeds per cell, frozen Po, the same pooled-std decision rule; only the manipulated factors and the primary axis differ (here precision, cond-IAE, is primary, because precision is the question).

The two factors, each built to remove a candidate constraint.

- **Teacher coverage (T).** A cascade **PID-CTBR** controller (gentle recovery gains) recovers the full **1–4 m band at 100 % survival, cond-IAE 0.14–0.18 m** ($\sim 2\times$ the 0.068 m oracle, $\sim 18\times$ tighter than the 2.8 m closed-loop floor) — a genuinely *competent* far-range teacher, where the state-PPO expert crashes beyond 2 m (§6.3, check 2). It collects far-range recovery demonstrations whose position-error mass lands **11.7 %** in 1–3 m ($p_{99} = 2.59$ m), versus the ~ 0.4 % of the hover data — a $\sim 29\times$ coverage jump, with zero crashes. **T₀** = hover-only data; **T₁** = hover + this far-range recovery set.
- **Observation (O).** **O₀** is the production saturating crosshair; **O₁** is the non-saturating perspective target of the §6.3 gate, now integrated into the production env, which restores far-range (≥ 1.5 m) image→distance R^2 from **0.05 to 0.40** (DR-on, in-env). **O₁** policies are *trained and evaluated* on the perspective observation.

Every cell is the **DoE1 frontier recipe** (Dispersive OFF, encoder E2E, H4-transfer init, task-conditioned, 500 hover + 500 recovery), so T and O are the only varying factors; the factors enter purely through which Phase-3 dataset is read. 12 BC runs (4 cells $\times$ seeds {0,1,2}), each scored through the *identical* `evaluate_frozen_p0` protocol (30 paired episodes, base seed 12345, $\sigma = 2.0$, $n_{\text{inf}} = 2$), with **O₁** cells handed their own perspective renderer at eval time (`evaluate_frozen_p0.py --target-render`).

Decision rule (pre-registered, mirroring §5). The floor is "broken" iff **all** hold: (1) `cond-IAE(T101) < cond-IAE(T000)` – pooled_std (significant vs the neither-factor control); (2) `cond-IAE(T101)  $\leq$  1.5 m` (an absolute target, $\sim 2\times$ off the floor); (3) `survival(T101)  $\geq$  survival(T000) – pooled_std` (a survival guard, because the §6.3 pos3d cue produced a "precision win" that was really a survival collapse); cond-IAE is trusted only when `n_cond  $\geq$  15` in both cells.

Table 6 — Teacher $\times$ Observation 2 $\times$ 2 (`evaluate_p2to_ablation.py`, 3 seeds, frozen protocol; `evaluation_results/p2to_ablation_leaderboard.json`; measured oracle 0.068 m, 0.9668 composite). cond-IAE is the **primary**

axis (lower is better); %Oracle is the secondary composite score, which rewards survival and precision jointly and so is not monotone in survival.

Cell	Teacher / Obs	cond-IAE (mean±std)	Survival	Tier-1	%Oracle	n_cond
ToOo	none / crosshair	2.69 ± 0.15 m (40× oracle)	83.2 ± 5.8 %	98.9 ± 1.6 %	20.1 %	29.7
ToO1	none / perspective	2.48 ± 0.14 m (36×)	76.7 ± 10.5 %	86.7 ± 9.8 %	22.1 %	26.0
T1Oo	far-range / crosshair	2.71 ± 0.22 m (40×)	91.4 ± 1.5 %	100.0 ± 0.0 %	23.1 %	30.0
T1O1	far-range / perspective	2.93 ± 0.18 m (43×)	90.6 ± 3.3 %	98.9 ± 1.6 %	20.8 %	29.7

Teacher × Observation 2×2 (3 seeds, frozen P0): FLOOR NOT BROKEN — T1O1–T0O0 = +0.24 m (pooled std 0.23 m), precision pinned, coverage×sensing interaction negative

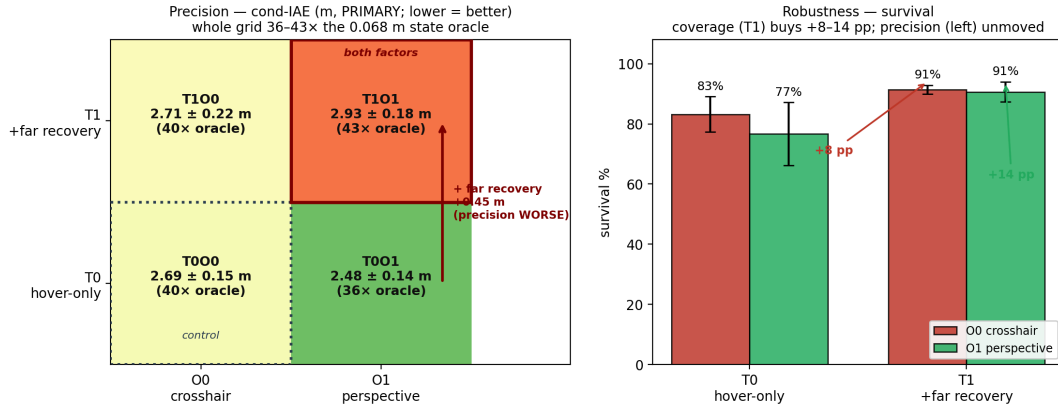


Figure 6. The Teacher × Observation 2×2 (3 seeds, mean ± across-seed std). *Left:* cond-IAE (primary; lower = more precise) as a 2×2 grid; the green dashed line is the 0.068 m state oracle and the whole grid sits at 36–43× it. *Right:* survival per cell. Coverage (T1, bottom row) lifts survival but not precision; the decisive T1O1-vs-ToOo contrast (arrow) is **+0.24 m — worse, and inside the 0.23 m pooled std — floor not broken**; the negative coverage×sensing interaction (ToO1 2.48 → T1O1 2.93 m) is annotated. (`scripts/make_paper_figures.py` → `evaluation_results/p2to_ablation_leaderboard.json`)

Verdict — floor NOT broken. The decisive T1O1-vs-ToOo comparison fails every breaking criterion: cond-IAE **2.93 vs 2.69 m** → $\Delta = +0.24$ m against a **pooled std of 0.23 m** (not significant — and in the *wrong* direction); 2.93 m $\gg$ the 1.5 m absolute target; only the survival guard passes (90.6 % $\geq$ 83.2 – 6.7 %). The floor holds across the **entire** 2×2 (2.48–2.93 m, 20–23 % oracle); $n_{\text{cond}} \geq 22$ everywhere, so this is not a short-survival artifact. **Even with the competent far-range teacher and the range-encoding observation supplied jointly, conditional hover precision does not move.**

What the grid shows — robustness ↑, precision pinned, a negative interaction. The result is not flat noise; it has clean, interpretable structure that *strengthens* the diagnosis (`experiment_report_p2to_decisive.md`):

- Coverage (T1) buys robustness, not precision.** Far-range recovery labels lift survival **+8.2 pp** on crosshair (83.2→91.4 %) and **+13.9 pp** on perspective (76.7→90.6 %) and pin Tier-1 at ~99–100 %, exactly the expected coverage benefit — yet move precision by **0** (T1Oo 2.71 m $\approx$ ToOo 2.69 m).
- Sensing (O1) is a weak precision lever and is not free.** On the hover-only row it improves cond-IAE 2.69→2.48 m ($\approx$ 0.21 m, $\sim 1\times$ the pooled std — marginal) while drifting survival 83.2→76.7 % and Tier-1 98.9→86.7 %.
- The interaction is NEGATIVE.** The single best precision cell is **ToO1 (2.48 m)**; adding far-range recovery on top of it (→ T1O1) *degrades* precision to **2.93 m (+0.45 m)**. The very factor that buys survival pulls the policy toward wide-range corrective behaviour that widens steady-state hover error.

This is a Robustness–Precision Capacity Conflict, made quantitative and RL-free. A v5 curriculum-RL run showed the same conflict but was confounded by RL dynamics (advantage masking); here it appears in a clean, three-seed **supervised** 2×2 whose decision rule was pre-registered before data collection (`RESEARCH_PLAN_v7.md`): within this policy's capacity one can have wide-range survival **or** tight hover precision, and pushing coverage trades the latter for the former. Two claim strengths should be kept distinct: the robustness–precision *conflict* is **directly observed** (the negative interaction is measured — survival-buying coverage costs precision, +0.45 m $>$ the 0.23 m pooled std), whereas *capacity* is the explanation we advance for it and then **test directly in §6.5**: scaling the action backbone 3.3× moves precision by at most ~ 0.3 m before saturating and does **not** break the floor, so capacity is at most a minor lever and the binding constraint is more fundamental than parameter count. The tension is the task-level signature of a phenomenon well documented in multi-task learning: competing objectives optimised under a fixed parameter budget trade off rather than co-improve — the multi-objective / Pareto view of Sener & Koltun [31], who note that a weighted sum of losses only works when tasks do not compete — and their gradients can directly conflict, so that a step improving one objective worsens the other (the negative-cosine "conflicting gradients" of Yu et al. [32]). Our coverage factor behaves as exactly such a competing objective: the gradient that buys survival is the one that worsens precision. Together with §6.1

(representation) and §6.3 (sensing), this is a **triple exclusion**: removing each of the three leading candidate constraints — separately and, for coverage × sensing, jointly — leaves the precision floor intact. The binding constraint is therefore **none of representation collapse, sensing, or coverage**, but a capacity / robustness–precision conflict.

6.5 Capacity is not the bottleneck: a `flow_net` scaling ladder

§6.4 leaves *capacity* as the one leading explanation we had not manipulated. We test it directly. The deployed policy carries **13.5 M** trainable parameters, 89 % of them in the `flow_net` action backbone. Holding the T1O1 recipe fixed — the cell where the conflict bites — and varying **only** that backbone's width and depth (`flow_net down_dims`), we train a three-rung ladder, 3 seeds each, scored under the identical frozen protocol on the perspective render.

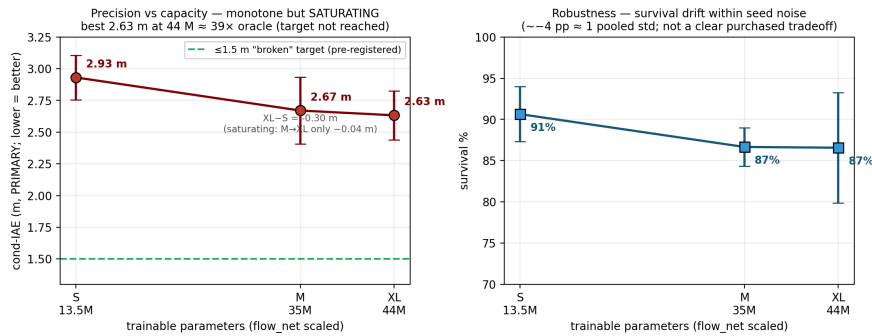
Table 7 — capacity scaling ladder (`evaluate_p2cap_ablation.py`, 3 seeds, frozen protocol, perspective render; `evaluation_results/p2cap_ablation_leaderboard.json`; measured oracle 0.068 m). cond-IAE is primary (lower = more precise); survival is the guard.

Rung	<code>flow_net down_dims</code>	trainable params	cond-IAE (m)	survival	Tier-1
S (baseline = T1O1)	(256, 512)	13.5 M	2.93 ± 0.18	90.6 %	98.9 %
M (deeper)	(256, 512, 768)	35.0 M	2.67 ± 0.26	86.7 %	100 %
XL (wider)	(512, 1024)	44.0 M	2.63 ± 0.19	86.6 %	94.4 %

Scaling the action backbone **3.3×** does not break the floor. cond-IAE improves monotonically (2.93 → 2.67 → 2.63 m) and the XL gain is statistically resolvable ($\Delta = -0.30 \text{ m} > \text{the } 0.26 \text{ m pooled std}$), but it is **small and saturating** (M→XL only -0.04 m) and stays at **~39× the 0.068 m oracle** ($\leq 24.5 \%$ oracle composite) — far from the pre-registered $\leq 1.5 \text{ m}$ target for a broken floor. Survival drifts down ~4 pp as capacity grows (90.6 → 86.6 %), but this drift is ~1 pooled std — within seed noise (the survival guard passes) — so we read it as *suggestive* of, not proof of, the precision gain being paid for in survival. Either way capacity does not dissolve the robustness–precision frontier; it at most nudges the operating point along it, and neither end is deployable.

Figure 7. The capacity ladder: cond-IAE (primary; *left*) and survival (*right*) vs trainable parameters (S 13.5 M, M 35 M, XL 44 M; 3 seeds, mean ± across-seed std). The green dashed line marks the pre-registered $\leq 1.5 \text{ m}$ "broken" target; cond-IAE descends monotonically but saturates well above it (best 2.63 m at 44 M, ~39× the 0.068 m oracle) while survival drifts down — capacity moves the operating point *along* the robustness–precision frontier without crossing the deployability line. (`make_paper_figures.py`, reads `evaluation_results/p2cap_ablation_leaderboard.json`.)

Capacity scaling ladder (3 seeds, frozen P0): FLOOR NOT BROKEN BY CAPACITY — scaling `flow_net` 3.3× (13.5M→44M) leaves cond-IAE ≈2.6 m (~39× oracle); capacity moves along the robustness–precision frontier, not across it



Two independent open-loop signals corroborate that 13.5 M already suffices to *fit* the data. First, all three rungs converge to the same best validation flow loss (≈ 0.0104) — 3.3× capacity does not lower the training objective. Second, an open-loop fitting probe (the Gate-A measurement that motivated this ladder) shows the 13.5 M generalist already fits both the hover and far-range distributions: its hover fit is only ~6 % worse than a hover specialist's (a real but tiny interference, 3.2× the seed std yet negligible in absolute terms) while its far-range fit is ~24× better. The small open-loop interference does not grow into a fitting-capacity wall, so the closed-loop floor is not a representational-capacity one.

Capacity is no longer the leading *untested* explanation: it is tested, and a 3.3× scale-up moves precision by at most ~0.3 m before saturating. The binding constraint is therefore *more fundamental than parameter count* — an inference from a 3.3× null, not a positive identification. Excluding raw parameter count leaves **two distinct candidates, both still untested and not to be conflated**: (a) **capacity allocation** — an architecture that need not spend a *single shared* budget on wide-range recovery to stay alive (e.g. a dedicated precision pathway); still a capacity/architecture story, but one a ladder varying *total* budget cannot reach. (b) **a closed-loop limit** — compounding error / partial observability under monocular FPV; not a capacity story at all. The ladder excludes the simplest reading, more parameters, and leaves (a) versus (b) as the next experiment; we advance neither as a result.

7. Discussion

What generalises. (a) The frozen protocol and the short-survival artifact it fixes are reusable for any survival-vs-precision visual control benchmark. (b) The "objective-gaming" pathology of a *scale-sensitive* dispersion regulariser is a general caution: a distance-maximising contrastive term can be trivially satisfied by norm inflation, *worsening* intrinsic rank ($\sim 287\times$ off-path, $\sim 9\times$ on-path; §6.1). We checked the two obvious escapes and both fail to rescue control. *Placement*: regularising the on-path feature the policy actually conditions on (§5) removes the auxiliary-only projection there is to inflate, yet still does not help (no gain when the encoder is trainable; mildly destabilising when frozen). *Criterion*: swapping in a **scale-invariant** form (unit-sphere InfoNCE or VICReg; §6.1) removes the norm-inflation gaming outright (feature norm $8.93\times \rightarrow \sim 1.3\times$) and produces *genuine* high-rank dispersion (effective rank 3.5 % $\rightarrow$ 75–85 % of dimensions) — the intended effect of the regulariser — yet closed-loop survival and precision still do not move (if anything cond-IAE regresses ~ 0.1 – 0.2 m). So neither the placement nor the criterion is what holds it back. (c) The rank-vs-survival decoupling — now established at both the off-path `vis_pooled` and the on-path `flow_net` mid-block, and across both scale-sensitive and scale-invariant criteria — argues that "prevent representation collapse" is not automatically the right lever for closed-loop control; the binding constraint must be located empirically.

Coverage and sensing are jointly insufficient. The §6.4 2×2 turns the §6.3 precision inference into a *test* and refutes it: supplying a competent far-range teacher (T1) and a range-encoding observation (O1) **together** leaves cond-IAE pinned at 2.93 m ($\approx 43\times$ oracle), no better than the neither-factor control. The grid also gives the robustness–precision trade-off its first clean, RL-free quantification on this task: coverage reliably buys **survival** (+8.2 pp crosshair, +13.9 pp perspective; Tier-1 $\rightarrow \sim 100$ %) but moves precision by 0, and the coverage $\times$ sensing **interaction is negative** — the best precision cell is T1O1 (2.48 m) and adding far-range recovery on top of it *worsens* precision to T1O1 (2.93 m). The factor that buys survival is the factor that costs precision. This reproduces, RL-free and pre-registered, the same conflict a v5 curriculum-RL run showed under confounded dynamics: within this capacity the policy can spend itself on wide-range survival **or** tight hover, not both — the robustness–precision instance of the capacity-bound, conflicting-objective trade-off long studied in multi-task learning [31, 32].

What is specific. The earlier reading — that the precision floor is specific to a teacher saturating inside the operating regime — does **not** survive §6.4. We removed that specific limitation: a PID-CTBR teacher competent across the full 1–4 m band (100 % recovery, cond-IAE 0.14–0.18 m) labelled the 1–3 m drift band densely, and precision still did not move. And capacity itself is *not* the specific limitation either: §6.5 scales the action backbone $3.3\times$ (13.5 M $\rightarrow$ 44 M trainable) under the identical protocol and the floor holds (best 2.63 m, $\sim 39\times$ oracle), the only gain a small, saturating one with at most a within-noise survival cost. What remains specific is the policy family, not the parameter count. Excluding raw capacity leaves two distinct untested candidates we do not conflate: *capacity allocation* — a dedicated precision pathway that need not spend a single shared budget on wide-range recovery (still an architecture story the ladder cannot reach) — and a *closed-loop* / partial-observability limit, which is not a capacity story at all.

Constructive implication. Improving metric precision here is neither a representation, a sensing, **nor** a coverage problem — the three fixes a reader would reach for first are each excluded (§6.1, §6.3, §6.4). The intervention shows even the *oracle* range barely moves precision and a richer cue collapses survival (§6.3); the decisive 2×2 shows that even adding a competent far-range teacher's coverage on top does not move it either, and trades precision for survival (§6.4); and *raw* capacity does not move it either — a $3.3\times$ larger action backbone leaves the floor at 2.63 m (§6.5). The remaining lever is therefore not more parameters but **capacity allocation and the closed-loop limit** — e.g. a dedicated precision pathway that need not spend a shared budget on wide-range recovery to stay alive, or a fix to the compounding error itself. Higher resolution, stereo/depth, representation regularisation, a $3.3\times$ larger backbone (§6.5), and wider BC coverage are all observed or predicted not to move the ~ 2.8 m floor on this architecture.

8. Limitations

- **Simulation only**; no real-robot validation. The renderer is synthetic and its distance encoding is a deliberate design choice (an information-poor monocular cue) — a feature for the information-gated analysis but a limit on external validity: the precision floor and its perspective-target remedy are established for this abstraction, not for photorealistic imagery. The §6.3 intervention controls for the renderer's information content (it hands the policy the oracle range directly), and the §6.4 teacher emits a CTBR command forward-compatible with a PX4 offboard interface, but validation under SITL/Gazebo rendering and on hardware (Jetson + PX4 MAVLink) remains future work.
- **The verdict is a closed-loop / capacity-allocation statement, not a teacher-competence or raw-capacity one.** One might expect the floor to be conditional on a teacher that saturates beyond 2 m; §6.4 removes that possibility by supplying a teacher competent across the full 1–4 m band (100 % recovery, cond-IAE 0.14–0.18 m) — precision still does not move. One might next expect it to be conditional on parameter count; §6.5 removes that too, by scaling the action backbone $3.3\times$ (13.5 M $\rightarrow$ 44 M trainable) under the identical protocol — the floor holds (best 2.63 m), with only a small, saturating precision gain and at most a within-noise survival cost. The verdict is instead conditional on the **policy family**: the robustness–precision conflict is established for this architecture (flow matching with IMU-vision cross-attention, 13.5 M

trainable) and is not relieved by more parameters. What was *not* varied are two distinct levers — capacity *allocation* (a dedicated precision pathway rather than a single shared backbone) and the closed-loop / partial-observability limit — which are the recommended next steps and out of scope for the pre-registered coverage $\times$ sensing question.

- **Data-volume asymmetry in the T $\times$ O 2 $\times$ 2.** To cells use hover-only data (500 ep), T1 cells hover + far-range recovery (1000 ep) — intrinsic to "add far-range labels". The asymmetry *favours* T1, yet precision does not improve under that favourable tilt, so it does not threaten the *negative* conclusion (it would matter only for a positive one). A size-matched near-recovery control is the fallback if a reviewer presses.
- **One task** (hover) and one policy family. The headline Dispersive null is established at the **official flow_net mid-block placement** ($\lambda=0.5$, $\tau=0.5$, $/d$); we additionally probe the off-path vis_pooled placement (§6.1) and, beyond the official InfoNCE-L2 form, two **scale-invariant** dispersion criteria (unit-sphere InfoNCE, VICReg; §6.1) that remove the objective-gaming yet still do not move control. We did not exhaustively sweep λ , τ , or every candidate intermediate block — though §6.1–§6.2 explain mechanistically why those are unlikely to rescue it here.
- **Statistical power.** Three seeds per cell — a pooled across-seed Tier-1 std of ~ 6.3 pp (Dispersive 2 $\times$ 2) and a pooled cond-IAE std of 0.23 m (T $\times$ O 2 $\times$ 2) — powers these designs to detect only large effects. Both results are nulls *in the sense of no effect above seed noise*; a smaller true effect, and the directional high-variance frozen-encoder degradation of §5, cannot be sharply resolved at this seed count.
- T_obs = 2 frames could in principle yield range via motion parallax; we argue (and the flat empirical precision across all configs supports) that the 28 ms baseline and slow drift make this far weaker than the static size channel.

9. Conclusion

On a vision-based quadrotor hover task, under a frozen, artifact-robust protocol with a measured oracle and an **official-code-faithful** Dispersive implementation (InfoNCE-L2 on the flow_net mid-block, $\lambda=0.5$, $/d$), **Dispersive Loss does not improve survival or precision** above seed noise (-2.2 pp Tier-1, $\in$ noise). Even with a frozen encoder — where the faithful term still trains the generative network — it does not help and is mildly destabilising, so the frozen-encoder condition is an informative control rather than a no-op. At **both** the faithful and an off-path placement the mechanism does not even cure the collapse it targets — it games its scale-sensitive objective and worsens intrinsic rank — and two **scale-invariant** criteria that *do* cure the collapse (effective rank to 75–85 % of dimensions, no norm inflation) still leave closed-loop control flat: survival is decoupled from representation geometry whatever criterion enforces it.

We then ran the diagnosis to its end. The survival gains come from a transfer/conditioning/recovery **recipe**, not Dispersive. And task precision — pinned at cond-IAE ≈ 2.8 m, ~ 13 % of the state oracle, across every configuration — is gated by **none** of the three leading candidates. A positive-control intervention rules out **sensing** (handing the policy the oracle metric position error barely moves precision, $\sim 36\times$ the state oracle; a richer cue collapses survival; the FPV's range-blindness is itself a fixable renderer artifact). A decisive, pre-registered **Teacher $\times$ Observation 2 $\times$ 2** then rules out **coverage**: a teacher competent across the full 1–4 m band and a range-encoding observation, supplied jointly, leave the floor at 2.93 m (no better than the neither-factor control), and coverage and sensing *negatively interact* — coverage buys survival but costs precision. The binding constraint for this task is therefore neither representation collapse, nor sensing, nor coverage, but a **robustness–precision conflict** — directly observed as that negative interaction. We then tested the leading explanation for that conflict, model capacity, by scaling the action backbone $3.3\times$ (13.5 M $\rightarrow$ 44 M trainable; §6.5): the floor held (best 2.63 m, $\sim 39\times$ oracle), the only gain small and saturating, with at most a within-noise survival cost. The conflict is therefore **more fundamental than parameter count** — leaving two distinct untested candidates we do not conflate: capacity *allocation* (a dedicated precision pathway rather than a single shared backbone) and a closed-loop / partial-observability limit under monocular FPV. We release the protocol, all three ablations (Dispersive $\times$ E2E, Teacher $\times$ Observation, capacity ladder), the intervention, and all diagnostics so the result and its diagnosis are reproducible.

Reproducibility / Artifacts

Component	Script	Artifact
Frozen protocol + measured oracle	<code>scripts/evaluate_frozen_p0.py</code>	<code>evaluation_results/frozen_p0_leaderboard.json</code>
§4 P1 baselines (BC-vision-only, PPO-from-pixels), 3 seeds	<code>scripts/train_bc_vision_only.py</code> , <code>scripts/train_ppo_from_pixels.py</code> (+ <code>models/ppo_pixel.py</code> , <code>configs/ppo_from_pixels.yaml</code>), <code>scripts/evaluate_baselines_frozen.py</code> ; seed driver <code>scripts/run_baseline_seeds_1_2.sh</code>	seed 0: <code>evaluation_results/baselines_frozen_leaderboard.json</code> , ... <code>_ppopx.json</code> ; seeds 1–2: <code>..._{bc,ppopx}_s{1,2}.json</code> ; aggregate (mean±std): <code>..._baselines_frozen_seeds_aggregate.json</code>
§5 faithful 2×2 (P2f, headline)	<code>scripts/run_p2_ablation.py --faithful</code> , <code>scripts/evaluate_p2_ablation.py</code>	<code>evaluation_results/p2f_ablation_{manifest,leaderboard}.json</code> — <code>docs/experiment_report_faithful_dispersive.md</code>
legacy 2×2 ablation (off-path, §6.1 probe)	<code>scripts/run_p2_ablation.py</code> , <code>scripts/evaluate_p2_ablation.py</code>	<code>evaluation_results/p2_ablation_{manifest,leaderboard}.json</code>
§6.1 feature geometry — off-path <code>vis_pooled</code> (Table 3)	<code>scripts/measure_feature_collapse.py</code>	<code>evaluation_results/p2_feature_collapse.json</code> — <code>docs/experiment_report_feature_collapse.md</code>
§6.1 feature geometry — faithful <code>flow_net</code> mid-block (Table 4)	<code>scripts/measure_feature_collapse_flowmid.py</code>	<code>evaluation_results/p2f_feature_collapse_flowmid.json</code>
§6.1 scale-invariant dispersion forms (Table 5)	<code>scripts/run_p6_form_ablation.py</code> (→ <code>train_flow_v5.py --dispersive-form {cosine,vicreg}</code>), <code>scripts/evaluate_p6_form_ablation.py</code>	<code>evaluation_results/p6_form_ablation_{manifest,leaderboard}.json</code> — <code>docs/experiment_report_p6_scale_invariant.md</code>
§6.2 survival movers	—	<code>docs/experiment_report_survival_movers.md</code>
§6.3 coverage probe	<code>scripts/measure_ood_coverage.py</code>	<code>evaluation_results/p3b_ood_coverage.json</code> — <code>docs/experiment_report_ood_coverage.md</code>
§6.3 image-distance info	<code>scripts/measure_image_distance_info.py</code>	<code>evaluation_results/p3b_image_distance_info{,_nodr}.json</code> — <code>docs/experiment_report_image_distance_info.md</code>
§6.3 higher-res gate (Fig 5 left)	<code>scripts/measure_higher_res_gate.py</code>	<code>evaluation_results/p3b_higher_res_gate.json</code> — <code>docs/experiment_report_sensing_ablation.md</code>
§6.3 range-cue intervention, 3 seeds (Fig 5 right)	<code>scripts/run_p3b_rangecue.py</code> (→ <code>train_flow_v5.py --range-cue</code> , <code>evaluate_frozen_p0.py --cue-noise</code>)	<code>evaluation_results/p3b_rc_{clean,noised}{,_s12}_frozen.json</code> — <code>docs/experiment_report_sensing_ablation.md</code>
§6.4 decisive Teacher × Observation 2×2 (Table 6, Fig 6)	<code>scripts/collect_data_v7_pidctr.py</code> , <code>scripts/run_p2to_ablation.py</code> (→ <code>train_flow_v5.py --hover-h5</code>), <code>scripts/evaluate_p2to_ablation.py</code> (→ <code>evaluate_frozen_p0.py --target-render</code>)	<code>evaluation_results/p2to_ablation_{manifest,leaderboard}.json</code> — <code>docs/experiment_report_p2to_decisive.md</code>
§6.5 capacity scaling ladder (Table 7, Fig 7)	<code>scripts/run_p2cap_ablation.py</code> (→ <code>train_flow_v5.py --down-dims</code>), <code>scripts/evaluate_p2cap_ablation.py</code> (capacity-agnostic <code>detect_arch</code>); open-loop Gate A <code>scripts/measure_capacity_fit.py</code>	<code>evaluation_results/p2cap_ablation_{manifest,leaderboard}.json</code> , <code>evaluation_results/p2cap_gateA_fit.json</code> — <code>docs/experiment_report_p2cap_capacity.md</code>

Frozen protocol: 30 episodes, base seed 12345, $\sigma = 2.0$ exp-decay composite, paired init, conditional-IAE over episodes surviving $\geq 250/500$ steps, bootstrap 95% CI, measured PPO oracle 0.9668.

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